

ERWIN POUSSI

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EDUCATION

Stanford University Stanford, CA	2025–2027 (expected)
M.S. in Aeronautics & Astronautics	GPA: 4.1/4.0
Coursework: <i>Robot Autonomy; Decision Making under Uncertainty; Optimal & Learning-Based Control; Deep RL.</i>	
École Polytechnique Palaiseau, France	2022–2025
M.Eng. in Mechanical Engineering; Autonomy concentration; Entrepreneurship minor.	GPA: 3.93/4.0
Lycée Blaise Pascal Preparatory program -, Mathematics & Physics; Clermont-Ferrand, France	2020–2022
Prytanée Militaire du Kadiogo (PMK) Military academy, Burkina Faso	2013–2020

RESEARCH & ENGINEERING

Stanford University Robotics Researcher	2026–Present
Multi-Robot Systems Lab & Autonomous Systems Lab — Drs. Mac Schwager and Marco Pavone	
• Designing legged-robot systems linking scene memory, grasping and motion planning for DARPA TIAMAT , a U.S. military initiative to rapidly deploy autonomous systems that adapt to changing environments. (project page)	
• Trained a language-conditioned visual drone-navigation policy using expert demonstrations and iterative imitation learning (DAGger), grounding instructions in visual targets and raising navigation success from 52% to 90% . (project page)	
Stanford School of Medicine Multi-Agent AI & Computational Biology Researcher , Qiu Lab	Oct. 2025–Present
• Formulated gene-panel selection as a reinforcement-learning problem, balancing panel size against cell-type discrimination; developing 3D/4D tissue-alignment tools for PantheonOS , a multi-agent AI framework for autonomous scientific research. (project page)	
• Modeling 3D embryo development and co-developing the Virtual Embryo Challenge’s agent interface for a shared benchmark; the broader project won one of eight seed grants in Laude Institute’s Moonshots and is competing for a \$10M AI-research lab . (project page)	
Stevens Institute of Technology Computational Fluid Dynamics & ML Intern	Apr.–Aug. 2025
• Built physics-informed, multi-fidelity surrogate models of Mars-landing parachute broadcloth, matching experimental permeability with <1% mean error ; this fabric type was also used in Orion’s Artemis II 2026 recovery parachutes . (project page)	
Daher Aerospace Multiphysics Simulation Intern	Jun.–Sep. 2024
• Developed and calibrated electromagnetic–thermal simulations of inductive composite welding in ANSYS, matching experimental temperatures to support control of heating and final part geometry.	
École Polytechnique / The Exploration Company Research Project	Sep. 2023–May 2024
• Developed vision-based navigation and guidance for Nyx’s autonomous docking; the spacecraft was selected by ESA for ISS cargo missions and pre-booked by Axiom Space for station resupply. (project page)	
French Police Cybercrime Unit Officer Cadet	Nov. 2022–Apr. 2023
• Led a team of trainee software engineers , working with police officers to develop an AI tool analyzing leaked communications, including the Conti leaks ; authored guides to share investigative methods with new unit members.	

COMMUNITY ENGAGEMENT

Stanford Black Graduate Student Association Fundraising Chair	Sep. 2025–Present
• Organized community events and raised \$3,000+ toward a \$10,000+ team fundraising effort supporting Black graduate students..	
Inspirit AI AI Instructor	Jul.–Aug. 2026
• Taught summer scholars to build machine-learning models and neural networks through hands-on programming.	
Cheer Up Outreach & External Relations	Oct. 2023–Apr. 2025
• Accompanied young cancer patients and coordinated outreach for a student-run cancer-support association.	
Mentoring & Opération Monge: Delivered presentations in underserved French high schools; provided one-to-one tutoring in mathematics and physics and university-application guidance to students in Burkina Faso, France and the U.S.	

PUBLICATIONS

Under review: Ghasimi, S. D., Poussi, E., Rabinovitch, J. “Surrogate-Based Optimization of Broadcloth Fabric Permeability for Mars Landing Parachute Simulations.” Submitted March 2026; reviews received August 2026; revision in progress.	
Preprint: Xu, W., Poussi, E., et al. “PantheonOS: An Evolvable Multi-Agent Framework for Automatic Genomics Discovery.” bioRxiv, 2026. doi:10.64898/2026.02.26.707870	

AWARDS & DISTINCTIONS

French Government Excellence Fellowship (2022) — French Embassy in Burkina Faso; awarded for École Polytechnique studies.	
Youth Olympiads, French Legion of Honor (SMLH) (2021) — 2nd place; team competition in culture and civic engagement.	
Burkina Faso baccalauréat (2020) — ranked 1st nationwide in mathematics & physical sciences (Series C); selected as spokesperson at the Presidency on behalf of students across Burkina Faso.	
PMK Model Student Award (2020) — Recognized academic achievement, exemplary conduct and school involvement.	
PMK Excellence Awards (2018–2020) — Highest academic score across all high-school grades, three consecutive years.	

SKILLS & INTERESTS

Technical: Python, C/C++, PyTorch, ROS2, MuJoCo, ANSYS; reinforcement learning, perception and control.	
Languages: French (native), English (TOEFL 109), Mooré (fluent), German (basic).	
Interests: Handball (French intercollegiate competition), strength training, drawing and painting; geopolitics in Africa.	